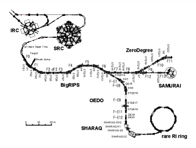
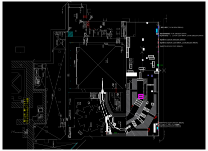
Archived External-Detector Review

Legacy H+V+V reference snapshot from 27 March 2026





ARCHIVED: not the current CompactInVacuum baseline and not for fabrication.

Current requirements and CAD are controlled by `docs/specs/compact_one_requirement_baseline.md` and `compactInVacuum/`.

Legacy External Polarimeter at afterSRC

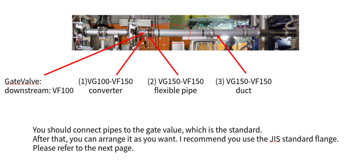
Role

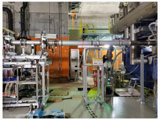
- Downstream polarimeter position after the SRC section.
- The polarimeter is organized around a dedicated target chamber.
- This archived concept used the external H+V+V detector arrangement.

Historical interface assumptions

- Front beam port: `ICF114`
- Rear beam port: `ICF114`
- Top rotary feedthrough mount: `ICF70`
- No extra side vacuum ports in this variant

Legacy External Polarimeter at the SAMURAI Side





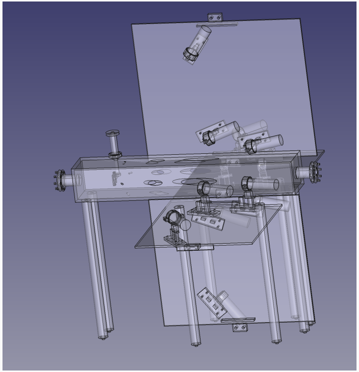
Role

- A SAMURAI-side polarimeter option is integration-driven rather than chamber-driven.
- The immediate boundary is defined by the beam pipe, the gate valve, and the local installation envelope.

Historical interface understanding

- Local standard family: **JIS** vacuum flanges and pipe-chain hardware
- Fixed gate-valve boundary at the local downstream side: `JIS VF100`
- Transition pipe (1): `JIS VG100--VF150` converter
- Transition pipe (2): `JIS VG150--VF150` flexible pipe
- Transition pipe (3): `JIS VG150--VF150` rigid duct
- Local space and maintenance envelope remain the dominant constraints for the upstream / in-front-of-SAMURAI polarimeter option

Archived afterSRC External Design



Historical interfaces visible in this view

- Front beamline interface: `ICF114` CF flange for the upstream beam connection
- Rear beamline interface: `ICF114` CF flange for the downstream beam connection
- Top target-drive interface: `ICF70` CF flange for the rotary feedthrough
- The chamber body is welded to short round beam pipes so the upstream and downstream beamline vacuum can be kept continuous after installation

Legacy layout note

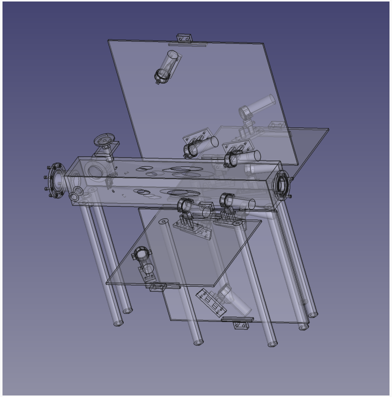
- This image belongs to the preserved external-detector reference route; it does not describe the current in-vacuum baseline

Fabrication-status correction

The LOS openings shown here were **temporary visualization cuts** in the archived model.

No wall-thinning or cutout statement from this deck is approved for fabrication. Any retained external-route chamber must be re-evaluated against the legacy baseline, vacuum boundary, external-pressure calculation, weld design, and current validation evidence.

Archived SAMURAI-front External Design



Historical interface interpretation

- The SAMURAI-side mechanical interface follows the existing **JIS** beam-pipe standard rather than the ICF chamber style used at afterSRC
- Fixed local beamline boundary at the gate valve: `JIS VF100`
- Required transition pieces in the present understanding are `JIS VG100--VF150`, then `JIS VG150--VF150` flexible and rigid sections

Mechanical consequence

- This option must be sized around the existing valve, pipe, and maintenance envelope close to SAMURAI
- Interface definition and available straight length remain the primary constraints for the final detector-side adapter geometry